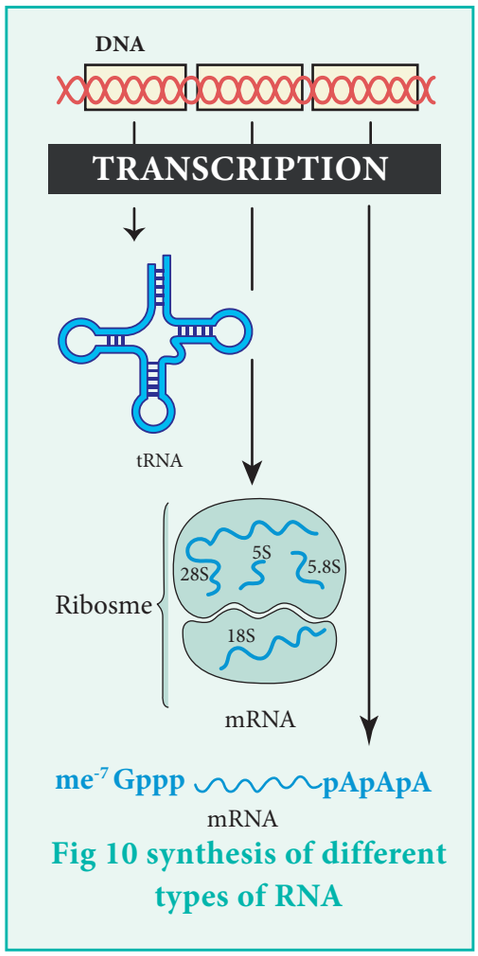


Fig 10 synthesis of different types of RNA

6.3.3.1 The Ribosomal RNA

These molecules are components of ribosomes. They form complexes with ribosomal proteins and act as important machinery for protein synthesis. There are three different species of rRNA, the 5S, 16S and 23S rRNA in prokaryotes and four in eukaryotes, the 5S, 5.8S, 18S and 28S rRNA. Here S represents the svedbergs unit or the sedimentation rate during centrifugation.

6.3.3.2 The Transfer RNA

These are the smallest among the three RNA types. Their primary function is to transfer aminoacids during protein synthesis. So there is atleast one tRNA molecule for every aminoacid. Figure 11 shows the secondary and tertiary structure of a typical tRNA molecule. The secondary structure is like a clover leaf and aminoacids are covalently linked to the 5'end. It could also be noted that there are many unusual bases in the tRNA like dihydrouracil. They also form extensive intra chain base pairing forming different loops in the RNA structure like the TψC loop, anticodon loop and D loop.

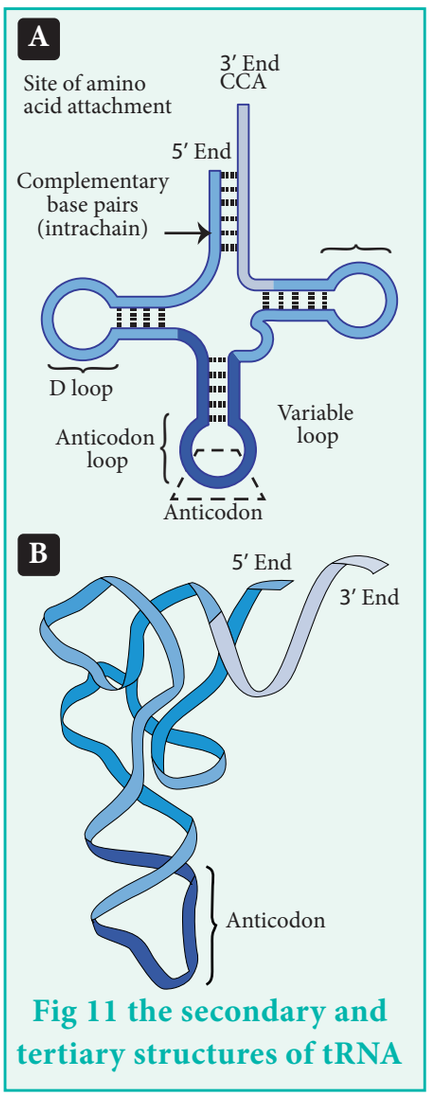


Fig 11 the secondary and tertiary structures of tRNA

6.3.3.3 The messenger RNA

This carries the genetic information from the DNA to the cytoplasam and hence the name messenger RNA. They are highly heterogeneous varying in size and sequence content. They act as template for protein synthesis. If the mRNA contains information for only one protein it is called monocistronic. The eukaryotic mRNAs are mostly monocistronic. However prokaryotic mRNA can contain information for encoding more than one mRNA and are called polycistronic.

6.3.4 Post transcriptional modification of RNA

In eukaryotes the RNA molecules are further processed. The most important modificatios are

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